



Mathematics AA 1.1 Arithmetic and geometric sequences and series State the formula for the nth term of a geometric sequence and explain each symbol. _____ IB-MATHAA-1.1-1 CONCEPT	Mathematics AA 1.1 Arithmetic and geometric sequences and series Why does the sum to infinity formula $S_{\infty} = \frac{u_1}{1-r}$ only apply for certain r values? _____ IB-MATHAA-1.1-2 CONCEPT	Mathematics AA 1.1 Arithmetic and geometric sequences and series An AP has $u_1=2$, $d=5$. What is u_{10} ? A 45 B 47 C 50 D 52 _____ IB-MATHAA-1.1-3 MULTIPLE CHOICE
Mathematics AA 1.1 Arithmetic and geometric sequences and series Find the sum of the first 6 terms of the GP 3,6,12,... _____ IB-MATHAA-1.1-4 PRACTICE	Mathematics AA 1.2 Exponents and logarithms State the change of base formula for $\log_a(x)$ in terms of natural logarithms. _____ IB-MATHAA-1.2-1 CONCEPT	Mathematics AA 1.2 Exponents and logarithms Why must the domain be checked after solving an equation involving $\log(x-2)$? _____ IB-MATHAA-1.2-2 CONCEPT
Mathematics AA 1.2 Exponents and logarithms $\log_5(125)$ equals: A 2 B 3 C 5 D 25 _____ IB-MATHAA-1.2-3 MULTIPLE CHOICE	Mathematics AA 1.2 Exponents and logarithms Solve $3^x = 50$ giving x to 3 s.f. _____ IB-MATHAA-1.2-4 PRACTICE	Mathematics AA 1.3 Financial applications of geometric sequences Write the compound interest formula and define each symbol. _____ IB-MATHAA-1.3-1 CONCEPT
Mathematics AA 1.3 Financial applications of geometric sequences Why is compound growth modelled by a geometric, not arithmetic, sequence? _____ IB-MATHAA-1.3-2 CONCEPT	Mathematics AA 1.3 Financial applications of geometric sequences \$1000 grows at 10% p.a. compounded annually. Value after 3 years? A \$1300 B \$1330 C \$1331 D \$1210 _____ IB-MATHAA-1.3-3 MULTIPLE CHOICE	Mathematics AA 1.3 Financial applications of geometric sequences Find the value of \$800 depreciating at 20% per year after 3 years. _____ IB-MATHAA-1.3-4 PRACTICE
Mathematics AA 1.4 The binomial theorem State the binomial theorem formula for $(a+b)^n$. _____ IB-MATHAA-1.4-1 CONCEPT	Mathematics AA 1.4 The binomial theorem How do you find a specific term (not the whole expansion) using the binomial theorem? _____ IB-MATHAA-1.4-2 CONCEPT	Mathematics AA 1.4 The binomial theorem The coefficient of x^2 in $(1+x)^4$ is: A 4 B 6 C 8 D 12 _____ IB-MATHAA-1.4-3 MULTIPLE CHOICE
Mathematics AA 1.4 The binomial theorem Find the coefficient of x^2 in $(2-x)^4$. _____ IB-MATHAA-1.4-4 PRACTICE	Mathematics AA 1.5 Complex numbers (HL) State De Moivre's theorem and what it is used for. _____ IB-MATHAA-1.5-1 CONCEPT	Mathematics AA 1.5 Complex numbers (HL) How do you find the modulus and argument of a complex number $a+bi$, and what care is needed for the argument? _____ IB-MATHAA-1.5-2 CONCEPT



Mathematics AA 1.5 Complex numbers (HL) (2+i)(1-i) equals: A 1-i B 3-i C 1+3i D 3+i IB-MATHAA-1.5-3 MULTIPLE CHOICE	Mathematics AA 1.5 Complex numbers (HL) Find $z_1 \times z_2$ in polar form given $z_1 = 2(\cos 30^\circ + i \sin 30^\circ)$, $z_2 = 3(\cos 40^\circ + i \sin 40^\circ)$. IB-MATHAA-1.5-4 PRACTICE	Mathematics AA 1.6 Proof by induction (HL) List the three components required in a complete proof by induction. IB-MATHAA-1.6-1 CONCEPT
Mathematics AA 1.6 Proof by induction (HL) Why is it insufficient to only check the base case and a few values of n? IB-MATHAA-1.6-2 CONCEPT	Mathematics AA 1.6 Proof by induction (HL) Induction is most suitable for proving statements about: A A single real number B All positive integers n C All real numbers x D Continuous functions only IB-MATHAA-1.6-3 MULTIPLE CHOICE	Mathematics AA 1.6 Proof by induction (HL) State the base case check needed to prove $2^n > n$ for all $n \geq 1$. IB-MATHAA-1.6-4 PRACTICE
Mathematics AA 1.7 Systems of linear equations (HL) Name the three possible solution types for a 3×3 linear system and their geometric meaning. IB-MATHAA-1.7-1 CONCEPT	Mathematics AA 1.7 Systems of linear equations (HL) What row operation is used to eliminate a variable between two equations in Gaussian elimination? IB-MATHAA-1.7-2 CONCEPT	Mathematics AA 1.7 Systems of linear equations (HL) Which row operation does NOT change the solution set of a linear system? A Multiplying a row by 0 B Swapping two rows C Deleting a row D Adding a constant to one side only IB-MATHAA-1.7-3 MULTIPLE CHOICE
Mathematics AA 1.7 Systems of linear equations (HL) Solve $x+y=5$, $2x-y=1$. IB-MATHAA-1.7-4 PRACTICE	Mathematics AA 2.1 Functions, domain and range What test determines whether a graph represents a function, and what does it check? IB-MATHAA-2.1-1 CONCEPT	Mathematics AA 2.1 Functions, domain and range Give two common reasons a value must be excluded from a function's domain. IB-MATHAA-2.1-2 CONCEPT
Mathematics AA 2.1 Functions, domain and range The range of $f(x)=x^2$ for domain $x \in \mathbb{R}$ is: A $f(x) \geq 0$ B $f(x) > 0$ C All real numbers D $f(x) \leq 0$ IB-MATHAA-2.1-3 MULTIPLE CHOICE	Mathematics AA 2.1 Functions, domain and range State the domain of $f(x)=\sqrt{x+4}$. IB-MATHAA-2.1-4 PRACTICE	Mathematics AA 2.2 Composite and inverse functions What is the geometric relationship between the graphs of $y=f(x)$ and $y=f^{-1}(x)$? IB-MATHAA-2.2-1 CONCEPT
Mathematics AA 2.2 Composite and inverse functions Why does a function need to be one-to-one (a bijection) for its inverse to exist as a function? IB-MATHAA-2.2-2 CONCEPT	Mathematics AA 2.2 Composite and inverse functions If $f(x)=x+5$, then $f^{-1}(x)$ is: A $x+5$ B $x-5$ C $1/(x+5)$ D $-x-5$ IB-MATHAA-2.2-3 MULTIPLE CHOICE	Mathematics AA 2.2 Composite and inverse functions If $f(x)=2x$ and $g(x)=x+3$, find $(f \circ g)(1)$. IB-MATHAA-2.2-4 PRACTICE



Mathematics AA 2.3 Quadratic functions and the discriminant	Mathematics AA 2.3 Quadratic functions and the discriminant	Mathematics AA 2.3 Quadratic functions and the discriminant
State the discriminant formula and what each sign of Δ tells you about the roots of $ax^2+bx+c=0$.	How do you find the vertex of a quadratic directly from its equation without completing the square?	The equation $x^2-4x+5=0$ has: A Two distinct real roots B One repeated real root C No real roots D Three real roots
IB-MATHAA-2.3-1CONCEPT	IB-MATHAA-2.3-2CONCEPT	IB-MATHAA-2.3-3MULTIPLE CHOICE
Mathematics AA 2.3 Quadratic functions and the discriminant	Mathematics AA 2.4 Transformations of graphs	Mathematics AA 2.4 Transformations of graphs
Find the vertex of $f(x)=x^2-2x+4$.	Describe the effect of the transformation $y=f(x)+d$ and $y=f(x+c)$ on the graph of $y=f(x)$.	What is the effect of a on the graph of $y=af(x)$ when $-1 < a < 0$?
IB-MATHAA-2.3-4PRACTICE	IB-MATHAA-2.4-1CONCEPT	IB-MATHAA-2.4-2CONCEPT
Mathematics AA 2.4 Transformations of graphs	Mathematics AA 2.4 Transformations of graphs	Mathematics AA 2.5 Exponential and logarithmic functions
$y=3f(x)$ transforms $y=f(x)$ by: A Horizontal stretch factor 3 B Vertical stretch factor 3 C Vertical translation of 3 D Horizontal translation of 3	If $y=f(x)$ has a maximum at (1,5), find the maximum point of $y=f(x)-2$.	State the domain, range, and asymptote of $y=\ln(x)$.
IB-MATHAA-2.4-3MULTIPLE CHOICE	IB-MATHAA-2.4-4PRACTICE	IB-MATHAA-2.5-1CONCEPT
Mathematics AA 2.5 Exponential and logarithmic functions	Mathematics AA 2.5 Exponential and logarithmic functions	Mathematics AA 2.5 Exponential and logarithmic functions
Why are $y=e^x$ and $y=\ln(x)$ reflections of each other in the line $y=x$?	The y-intercept of $y=4^x + 1$ is: A 0 B 1 C 2 D 4	State the horizontal asymptote of $y=7(0.5)^x - 4$.
IB-MATHAA-2.5-2CONCEPT	IB-MATHAA-2.5-3MULTIPLE CHOICE	IB-MATHAA-2.5-4PRACTICE
Mathematics AA 2.6 Rational functions (HL)	Mathematics AA 2.6 Rational functions (HL)	Mathematics AA 2.6 Rational functions (HL)
For $f(x)=(ax+b)/(cx+d)$, state how to find the vertical and horizontal asymptotes.	Why does a rational function $(ax+b)/(cx+d)$ never actually reach its horizontal asymptote?	The vertical asymptote of $f(x)=(x+2)/(2x-6)$ is: A $x=6$ B $x=3$ C $x=-2$ D $x=2$
IB-MATHAA-2.6-1CONCEPT	IB-MATHAA-2.6-2CONCEPT	IB-MATHAA-2.6-3MULTIPLE CHOICE
Mathematics AA 2.6 Rational functions (HL)	Mathematics AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics AA 2.7 Polynomial functions and factor theorem (HL)
State the horizontal asymptote of $f(x)=(4x-1)/(2x+5)$.	State the factor theorem.	How does the remainder theorem differ from the factor theorem?
IB-MATHAA-2.6-4PRACTICE	IB-MATHAA-2.7-1CONCEPT	IB-MATHAA-2.7-2CONCEPT
Mathematics AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics AA 3.1 Volume and surface area of 3D solids
$P(x)=x^3-x-6$ has $P(2)=0$. Which is a factor of $P(x)$? A $(x-6)$ B $(x+2)$ C $(x-2)$ D $(x+6)$	Find the remainder when x^3-2x+1 is divided by $(x-2)$.	State the volume formula for a cone and for a sphere.
IB-MATHAA-2.7-3MULTIPLE CHOICE	IB-MATHAA-2.7-4PRACTICE	IB-MATHAA-3.1-1CONCEPT



Mathematics AA 3.1 Volume and surface area of 3D solids	Mathematics AA 3.1 Volume and surface area of 3D solids	Mathematics AA 3.1 Volume and surface area of 3D solids
What is the slant height of a cone and how is it calculated?	The curved surface area of a cylinder with $r=2$, $h=7$ is: A 14π B 28π C 44π D 56π	Find the volume of a cuboid with dimensions 3 cm $\times$ 4 cm $\times$ 5 cm.
IB-MATHAA-3.1-2 CONCEPT	IB-MATHAA-3.1-3 MULTIPLE CHOICE	IB-MATHAA-3.1-4 PRACTICE
Mathematics AA 3.2 Right-angled trigonometry and bearings	Mathematics AA 3.2 Right-angled trigonometry and bearings	Mathematics AA 3.2 Right-angled trigonometry and bearings
State the SOH-CAH-TOA ratios for a right-angled triangle.	How are bearings measured, and what is the rule for expressing them?	The angle of elevation and angle of depression between two points are: A Always different B Equal (alternate angles) C Complementary D Supplementary
IB-MATHAA-3.2-1 CONCEPT	IB-MATHAA-3.2-2 CONCEPT	IB-MATHAA-3.2-3 MULTIPLE CHOICE
Mathematics AA 3.2 Right-angled trigonometry and bearings	Mathematics AA 3.3 Non-right triangles: sine and cosine rules	Mathematics AA 3.3 Non-right triangles: sine and cosine rules
A ladder of length 6 m leans against a wall making 60° with the ground. How high up the wall does it reach?	State the sine rule and the cosine rule (side form).	When does the 'ambiguous case' arise when using the sine rule, and how do you check for it?
IB-MATHAA-3.2-4 PRACTICE	IB-MATHAA-3.3-1 CONCEPT	IB-MATHAA-3.3-2 CONCEPT
Mathematics AA 3.3 Non-right triangles: sine and cosine rules	Mathematics AA 3.3 Non-right triangles: sine and cosine rules	Mathematics AA 3.4 The unit circle and radian measure
The area of a triangle with sides 6 and 8 and included angle 90° is: A 24 B 48 C 14 D 30	In triangle ABC, $a=6$, $b=6$, $C=60^\circ$. Find c.	State the formulas for arc length and sector area in terms of radius r and angle θ in radians.
IB-MATHAA-3.3-3 MULTIPLE CHOICE	IB-MATHAA-3.3-4 PRACTICE	IB-MATHAA-3.4-1 CONCEPT
Mathematics AA 3.4 The unit circle and radian measure	Mathematics AA 3.4 The unit circle and radian measure	Mathematics AA 3.4 The unit circle and radian measure
On the unit circle, how are $\cos\theta$ and $\sin\theta$ defined geometrically?	$\pi/3$ radians is equivalent to: A 30° B 45° C 60° D 90°	Find the arc length of a sector with radius 5 cm and angle $\pi/2$ radians.
IB-MATHAA-3.4-2 CONCEPT	IB-MATHAA-3.4-3 MULTIPLE CHOICE	IB-MATHAA-3.4-4 PRACTICE
Mathematics AA 3.5 Trigonometric identities and equations	Mathematics AA 3.5 Trigonometric identities and equations	Mathematics AA 3.5 Trigonometric identities and equations
State the Pythagorean identity and one double angle formula for cosine.	When solving a trig equation like $\sin(2x)=k$ over a domain, why must you find ALL solutions of the argument before dividing back by 2?	$\sin^2x + \cos^2x$ is always equal to: A 0 B 1 C $2\sin x \cos x$ D $\tan^2x$
IB-MATHAA-3.5-1 CONCEPT	IB-MATHAA-3.5-2 CONCEPT	IB-MATHAA-3.5-3 MULTIPLE CHOICE



Mathematics AA 3.5 Trigonometric identities and equations	Mathematics AA 3.6 Vectors and scalar/vector products (HL)	Mathematics AA 3.6 Vectors and scalar/vector products (HL)
Solve $\sin x = 1/2$ for $0 \leq x \leq 2\pi$.	State the formula for the scalar (dot) product of two vectors in terms of their magnitudes and the angle between them.	What geometric quantity does $ a \times b $ represent, and how is it used to find a triangle's area?
IB-MATHAA-3.5-4PRACTICE	IB-MATHAA-3.6-1CONCEPT	IB-MATHAA-3.6-2CONCEPT
Mathematics AA 3.6 Vectors and scalar/vector products (HL)	Mathematics AA 3.6 Vectors and scalar/vector products (HL)	Mathematics AA 3.7 Lines and planes in three dimensions (HL)
For $a=(1,0,0)$ and $b=(0,1,0)$, $a \cdot b$ equals: A 1 B 0 C -1 D (0,0,1)	Find $ v $ for $v=(3,4,0)$.	Write the vector equation of a line and the equation of a plane, defining all symbols.
IB-MATHAA-3.6-3MULTIPLE CHOICE	IB-MATHAA-3.6-4PRACTICE	IB-MATHAA-3.7-1CONCEPT
Mathematics AA 3.7 Lines and planes in three dimensions (HL)	Mathematics AA 3.7 Lines and planes in three dimensions (HL)	Mathematics AA 3.7 Lines and planes in three dimensions (HL)
How do you find the angle between a line and a plane, given the line's direction vector and the plane's normal vector?	Two planes with normals $n_1=(1,0,0)$ and $n_2=(0,1,0)$ are: A Parallel B Perpendicular C Identical D Intersecting at an obtuse angle	Find the point where the line $r=(0,0,0)+t(1,1,1)$ meets the plane $x+y+z=6$.
IB-MATHAA-3.7-2CONCEPT	IB-MATHAA-3.7-3MULTIPLE CHOICE	IB-MATHAA-3.7-4PRACTICE
Mathematics AA 4.1 Sampling and data collection	Mathematics AA 4.1 Sampling and data collection	Mathematics AA 4.1 Sampling and data collection
Describe stratified sampling and why it is used.	What is sampling bias, and give one cause of it.	Which method gives every member of the population an equal chance of selection? A Convenience sampling B Quota sampling C Simple random sampling D Systematic sampling
IB-MATHAA-4.1-1CONCEPT	IB-MATHAA-4.1-2CONCEPT	IB-MATHAA-4.1-3MULTIPLE CHOICE
Mathematics AA 4.1 Sampling and data collection	Mathematics AA 4.2 Statistical measures and displays	Mathematics AA 4.2 Statistical measures and displays
A population has 400 men and 600 women. Find the number of women in a stratified sample of size 50.	State the formula for the interquartile range and the rule for identifying an outlier using it.	Why is the median often preferred over the mean when a data set contains outliers?
IB-MATHAA-4.1-4PRACTICE	IB-MATHAA-4.2-1CONCEPT	IB-MATHAA-4.2-2CONCEPT
Mathematics AA 4.2 Statistical measures and displays	Mathematics AA 4.2 Statistical measures and displays	Mathematics AA 4.3 Correlation and linear regression
A box plot's central line represents the: A Mean B Mode C Median D Range	Find the range and IQR of the data set: 3, 5, 5, 8, 9, 11, 14.	What does Pearson's correlation coefficient r measure, and what range of values can it take?
IB-MATHAA-4.2-3MULTIPLE CHOICE	IB-MATHAA-4.2-4PRACTICE	IB-MATHAA-4.3-1CONCEPT



Mathematics AA 4.3 Correlation and linear regression	Mathematics AA 4.3 Correlation and linear regression	Mathematics AA 4.3 Correlation and linear regression
Why is extrapolation using a regression line considered unreliable?	Which value of r shows the weakest linear correlation? A $r=0.95$ B $r=-0.88$ C $r=0.05$ D $r=-1.0$	A regression line is $y=3x+2$. Estimate y when $x=4$.
IB-MATHAA-4.3-2 CONCEPT	IB-MATHAA-4.3-3 MULTIPLE CHOICE	IB-MATHAA-4.3-4 PRACTICE
Mathematics AA 4.4 Probability, Venn and tree diagrams	Mathematics AA 4.4 Probability, Venn and tree diagrams	Mathematics AA 4.4 Probability, Venn and tree diagrams
State the addition rule for $P(A \cup B)$ and how it simplifies for mutually exclusive events.	State the condition for two events A and B to be independent, in terms of probabilities.	$P(A)=0.6$, $P(B)=0.3$, $P(A \cap B)=0.18$. Are A and B independent? A Yes, since $0.6 \times 0.3 = 0.18$ B No, since $P(A \cap B) \neq 0$ C Yes, since $P(A \cup B) > 0$ D Cannot be determined
IB-MATHAA-4.4-1 CONCEPT	IB-MATHAA-4.4-2 CONCEPT	IB-MATHAA-4.4-3 MULTIPLE CHOICE
Mathematics AA 4.4 Probability, Venn and tree diagrams	Mathematics AA 4.5 Discrete random variables and the binomial distribution	Mathematics AA 4.5 Discrete random variables and the binomial distribution
$P(A)=0.4$, $P(B)=0.5$, A and B mutually exclusive. Find $P(A \cup B)$.	State the formulas for the mean and variance of a binomial distribution $X \sim B(n, p)$.	What conditions must be satisfied for a random variable to be modelled by a binomial distribution?
IB-MATHAA-4.4-4 PRACTICE	IB-MATHAA-4.5-1 CONCEPT	IB-MATHAA-4.5-2 CONCEPT
Mathematics AA 4.5 Discrete random variables and the binomial distribution	Mathematics AA 4.5 Discrete random variables and the binomial distribution	Mathematics AA 4.6 The normal distribution
For a discrete random variable, $\sum P(X=x)$ over all values of x must equal: A 0 B 0.5 C 1 D n	$X \sim B(5, 0.5)$. Find $E(X)$ and $\text{Var}(X)$.	State the standardisation formula for converting a normal variable X to the standard normal Z .
IB-MATHAA-4.5-3 MULTIPLE CHOICE	IB-MATHAA-4.5-4 PRACTICE	IB-MATHAA-4.6-1 CONCEPT
Mathematics AA 4.6 The normal distribution	Mathematics AA 4.6 The normal distribution	Mathematics AA 4.6 The normal distribution
State the approximate percentages of data within 1, 2, and 3 standard deviations of the mean in a normal distribution.	For $X \sim N(\mu, \sigma^2)$, $P(X < \mu)$ equals: A 0 B 0.25 C 0.5 D 1	$X \sim N(20, 5^2)$. Find the z-score for $x=25$.
IB-MATHAA-4.6-2 CONCEPT	IB-MATHAA-4.6-3 MULTIPLE CHOICE	IB-MATHAA-4.6-4 PRACTICE
Mathematics AA 4.7 Bayes' theorem (HL)	Mathematics AA 4.7 Bayes' theorem (HL)	Mathematics AA 4.7 Bayes' theorem (HL)
State Bayes' theorem for $P(A B)$.	How is the law of total probability used within Bayes' theorem?	In Bayes' theorem, $P(A)$ (before observing B) is called the: A Posterior probability B Prior probability C Conditional probability D Joint probability
IB-MATHAA-4.7-1 CONCEPT	IB-MATHAA-4.7-2 CONCEPT	IB-MATHAA-4.7-3 MULTIPLE CHOICE



Mathematics AA 4.7 Bayes' theorem (HL) $P(B A)=0.8$, $P(A)=0.25$, $P(B)=0.4$. Find $P(A B)$. IB-MATHAA-4.7-4 PRACTICE	Mathematics AA 5.1 Limits and the derivative from first principles State the definition of the derivative of $f(x)$ from first principles. IB-MATHAA-5.1-1 CONCEPT	Mathematics AA 5.1 Limits and the derivative from first principles What does the derivative $f'(x)$ represent geometrically? IB-MATHAA-5.1-2 CONCEPT
Mathematics AA 5.1 Limits and the derivative from first principles For $f(x)=5$ (a constant), $f'(x)$ from first principles equals: A 5 B 0 C x D Undefined IB-MATHAA-5.1-3 MULTIPLE CHOICE	Mathematics AA 5.1 Limits and the derivative from first principles Using first principles, find $f'(x)$ for $f(x)=4x$. IB-MATHAA-5.1-4 PRACTICE	Mathematics AA 5.2 Differentiation rules and tangents/normals State the product rule and the quotient rule for differentiation. IB-MATHAA-5.2-1 CONCEPT
Mathematics AA 5.2 Differentiation rules and tangents/normals How is the gradient of the normal to a curve at a point related to the gradient of the tangent there? IB-MATHAA-5.2-2 CONCEPT	Mathematics AA 5.2 Differentiation rules and tangents/normals $d/dx(x^5)$ equals: A x^4 B $5x^6$ C $5x^4$ D $x^6/6$ IB-MATHAA-5.2-3 MULTIPLE CHOICE	Mathematics AA 5.2 Differentiation rules and tangents/normals Find dy/dx for $y=x^2+3x$ at $x=2$. IB-MATHAA-5.2-4 PRACTICE
Mathematics AA 5.3 Optimisation and related rates How do you classify a critical point $x=a$ (where $f'(a)=0$) using the second derivative test? IB-MATHAA-5.3-1 CONCEPT	Mathematics AA 5.3 Optimisation and related rates In a related rates problem, what role does the chain rule play? IB-MATHAA-5.3-2 CONCEPT	Mathematics AA 5.3 Optimisation and related rates For $f(x)=x^2-6x+5$, the minimum value occurs at: A $x=3$ B $x=6$ C $x=-3$ D $x=0$ IB-MATHAA-5.3-3 MULTIPLE CHOICE
Mathematics AA 5.3 Optimisation and related rates Find the critical point of $f(x)=x^2-4x+1$ and state whether it is a maximum or minimum. IB-MATHAA-5.3-4 PRACTICE	Mathematics AA 5.4 Kinematics using calculus State the relationships between displacement, velocity, and acceleration using calculus notation. IB-MATHAA-5.4-1 CONCEPT	Mathematics AA 5.4 Kinematics using calculus How do you determine when a moving particle is momentarily at rest? IB-MATHAA-5.4-2 CONCEPT
Mathematics AA 5.4 Kinematics using calculus If $a(t)=4$ (constant) and $v(0)=2$, what is $v(t)$? A $4t$ B $4t+2$ C $2t+4$ D $4t-2$ IB-MATHAA-5.4-3 MULTIPLE CHOICE	Mathematics AA 5.4 Kinematics using calculus A particle has $s(t)=t^2-4t$. Find its velocity at $t=3$. IB-MATHAA-5.4-4 PRACTICE	Mathematics AA 5.5 Integration and areas under curves State the power rule for integration and one standard exponential/logarithmic integral. IB-MATHAA-5.5-1 CONCEPT



Mathematics AA 5.5 Integration and areas under curves Why must you check which curve is above the other before computing the area between two curves? IB-MATHAA-5.5-2 CONCEPT	Mathematics AA 5.5 Integration and areas under curves $\int [1 \text{ to } 3] 2x \, dx$ equals: A 4 B 6 C 8 D 10 IB-MATHAA-5.5-3 MULTIPLE CHOICE	Mathematics AA 5.5 Integration and areas under curves Find $\int (3x^2-2)dx$. IB-MATHAA-5.5-4 PRACTICE
Mathematics AA 5.6 Volumes of revolution (HL) State the formula for the volume of revolution when $y=f(x)$ is rotated 360° about the x-axis between $x=a$ and $x=b$. IB-MATHAA-5.6-1 CONCEPT	Mathematics AA 5.6 Volumes of revolution (HL) Why does the volume of revolution formula involve y^2 rather than y ? IB-MATHAA-5.6-2 CONCEPT	Mathematics AA 5.6 Volumes of revolution (HL) The region under $y=2$ (constant) from $x=0$ to $x=5$, rotated about the x-axis, gives volume: A 10π B 20π C 25π D 50π IB-MATHAA-5.6-3 MULTIPLE CHOICE
Mathematics AA 5.6 Volumes of revolution (HL) Set up (do not evaluate) the integral for the volume when $y=x^2$ ($0 \leq x \leq 1$) is rotated about the x-axis. IB-MATHAA-5.6-4 PRACTICE	Mathematics AA 5.7 Further differentiation: implicit and higher derivatives (HL) When differentiating an equation implicitly, how do you handle a term like y^2 or xy with respect to x ? IB-MATHAA-5.7-1 CONCEPT	Mathematics AA 5.7 Further differentiation: implicit and higher derivatives (HL) Why is implicit differentiation necessary for curves like $x^2+y^2=25$? IB-MATHAA-5.7-2 CONCEPT
Mathematics AA 5.7 Further differentiation: implicit and higher derivatives (HL) For $x+y^2=10$, dy/dx equals: A $-1/(2y)$ B $1/(2y)$ C $-2y$ D $2y$ IB-MATHAA-5.7-3 MULTIPLE CHOICE	Mathematics AA 5.7 Further differentiation: implicit and higher derivatives (HL) Find dy/dx for the curve $x^2-y=4$. IB-MATHAA-5.7-4 PRACTICE	Mathematics AA 5.8 Further integration: substitution and by parts (HL) State the integration by parts formula. IB-MATHAA-5.8-1 CONCEPT
Mathematics AA 5.8 Further integration: substitution and by parts (HL) When using substitution $u=g(x)$, what must you remember to convert besides the function itself? IB-MATHAA-5.8-2 CONCEPT	Mathematics AA 5.8 Further integration: substitution and by parts (HL) For $\int 3x^2(x^3+1)^5 \, dx$, a suitable substitution is: A $u=x^2$ B $u=x^3+1$ C $u=x^5$ D $u=3x^2$ IB-MATHAA-5.8-3 MULTIPLE CHOICE	Mathematics AA 5.8 Further integration: substitution and by parts (HL) Find $\int 2x \cdot \cos(x^2)dx$ using substitution. IB-MATHAA-5.8-4 PRACTICE
Mathematics AA 5.9 Differential equations and Maclaurin series (HL) Outline the method for solving a separable differential equation $dy/dx=f(x)g(y)$. IB-MATHAA-5.9-1 CONCEPT	Mathematics AA 5.9 Differential equations and Maclaurin series (HL) State the general Maclaurin series formula for $f(x)$ about $x=0$. IB-MATHAA-5.9-2 CONCEPT	Mathematics AA 5.9 Differential equations and Maclaurin series (HL) For the Maclaurin series of $\sin(x)$, the coefficient of x^3 is: A $1/6$ B $-1/6$ C $1/3$ D $-1/3$ IB-MATHAA-5.9-3 MULTIPLE CHOICE

**Mathematics AA**

5.9 Differential equations and Maclaurin series (HL)

Solve $\frac{dy}{dx} = 2y$ (separable), given
 $y=5$ when $x=0$.

IB-MATHAA-5.9-4

PRACTICE



Mathematics AA

1.1 Arithmetic and geometric sequences and series

B. $u_{10}=2+9(5)=47$.

IB-MATHAA-1.1-3

ANSWER

Mathematics AA

1.1 Arithmetic and geometric sequences and series

Only when $|r|<1$, since then $r^n \rightarrow 0$ as $n \rightarrow \infty$ so the partial sums converge to a finite limit; otherwise the series diverges.

IB-MATHAA-1.1-2

ANSWER

Mathematics AA

1.1 Arithmetic and geometric sequences and series

$u_n = u_1 \cdot r^{(n-1)}$, where u_1 is the first term, r the common ratio, and n the term number.

IB-MATHAA-1.1-1

ANSWER

Mathematics AA

1.2 Exponents and logarithms

Because logarithms are only defined for positive arguments, so any solution making $x-2 \leq 0$ (or the other argument ≤ 0) must be rejected.

IB-MATHAA-1.2-2

ANSWER

Mathematics AA

1.2 Exponents and logarithms

$\log_a(x) = \ln(x)/\ln(a)$.

IB-MATHAA-1.2-1

ANSWER

Mathematics AA

1.1 Arithmetic and geometric sequences and series

$r=2$,
 $S_n=u_1(r^n-1)/(r-1)=3(2^6-1)/1=3(63)=189$.

IB-MATHAA-1.1-4

ANSWER

Mathematics AA

1.3 Financial applications of geometric sequences

FV = $P(1+r/(100k))^{(kn)}$: P principal, r nominal annual rate (%), k compounding periods per year, n number of years.

IB-MATHAA-1.3-1

ANSWER

Mathematics AA

1.2 Exponents and logarithms

$x = \ln 50 / \ln 3 = 3.9120 / 1.0986 = 3.56$.

IB-MATHAA-1.2-4

ANSWER

Mathematics AA

1.2 Exponents and logarithms

B. $5^3=125$, so $\log_5(125)=3$.

IB-MATHAA-1.2-3

ANSWER

Mathematics AA

1.3 Financial applications of geometric sequences

$800 \times 0.8^3 = 800 \times 0.512 = \409.60 .

IB-MATHAA-1.3-4

ANSWER

Mathematics AA

1.3 Financial applications of geometric sequences

C. $1000(1.1)^3 = 1331$.

IB-MATHAA-1.3-3

ANSWER

Mathematics AA

1.3 Financial applications of geometric sequences

Because each compounding period multiplies the current value by a constant factor $(1+r/(100k))$, giving a common ratio rather than a common difference.

IB-MATHAA-1.3-2

ANSWER

Mathematics AA

1.4 The binomial theorem

B. $4C_2 = 6$.

IB-MATHAA-1.4-3

ANSWER

Mathematics AA

1.4 The binomial theorem

Use the general term $nCr \cdot a^{(n-r)} \cdot b^r$ and solve for the value of r that gives the required power of the variable, then substitute.

IB-MATHAA-1.4-2

ANSWER

Mathematics AA

1.4 The binomial theorem

$(a+b)^n = \sum_{r=0}^n nCr \cdot a^{(n-r)} \cdot b^r$.

IB-MATHAA-1.4-1

ANSWER

Mathematics AA

1.5 Complex numbers (HL)

$r=\sqrt{a^2+b^2}$; $\theta=\arctan(b/a)$ adjusted for the correct quadrant based on the signs of a and b , since $\arctan$ alone only gives values in $(-\pi/2, \pi/2)$.

IB-MATHAA-1.5-2

ANSWER

Mathematics AA

1.5 Complex numbers (HL)

$z^n = r^n(\cos n\theta + i \sin n\theta)$ for $z=r(\cos\theta+i\sin\theta)$; used to compute powers of complex numbers and to find n th roots.

IB-MATHAA-1.5-1

ANSWER

Mathematics AA

1.4 The binomial theorem

General term: $4Cr \cdot 2^{(4-r)} \cdot (-x)^r$;
 $r=2$: $4C_2 \cdot 2^{2 \cdot 1} = 6 \times 4 = 24$.

IB-MATHAA-1.4-4

ANSWER

Mathematics AA

1.6 Proof by induction (HL)

1) Base case verification, 2) Inductive step: assume $P(k)$ true, prove $P(k+1)$ true, 3) Conclusion stating the principle of mathematical induction applies for all $n \geq n_0$.

IB-MATHAA-1.6-1

ANSWER

Mathematics AA

1.5 Complex numbers (HL)

Moduli multiply, arguments add:
 $z_1 z_2 = 6(\cos 70^\circ + i \sin 70^\circ)$.

IB-MATHAA-1.5-4

ANSWER

Mathematics AA

1.5 Complex numbers (HL)

B. $(2+i)(1-i) = 2-2i+i-i^2 = 2-i+1 = 3-i$.

IB-MATHAA-1.5-3

ANSWER



Mathematics AA 1.6 Proof by induction (HL)	Mathematics AA 1.6 Proof by induction (HL)	Mathematics AA 1.6 Proof by induction (HL)
Check $n=1$: $2^1=2 > 1$, true, so the base case holds.	B. Induction applies to statements indexed by integers, typically all $n \geq$ some starting value.	Because checking finitely many cases does not prove the statement for all n ; the inductive step is essential to show truth propagates from any k to $k+1$, covering the infinite set of remaining cases.
IB-MATHAA-1.6-4ANSWER	IB-MATHAA-1.6-3ANSWER	IB-MATHAA-1.6-2ANSWER
Mathematics AA 1.7 Systems of linear equations (HL)	Mathematics AA 1.7 Systems of linear equations (HL)	Mathematics AA 1.7 Systems of linear equations (HL)
B. Swapping rows reorders equations without altering the solution set; the other options can change or lose information.	Add or subtract a suitable multiple of one equation (row) to/from another so that the coefficient of the chosen variable becomes zero.	Unique solution (three planes meet at one point), no solution (inconsistent, e.g. parallel/skew planes with no common point), infinitely many solutions (planes intersect in a common line).
IB-MATHAA-1.7-3ANSWER	IB-MATHAA-1.7-2ANSWER	IB-MATHAA-1.7-1ANSWER
Mathematics AA 2.1 Functions, domain and range	Mathematics AA 2.1 Functions, domain and range	Mathematics AA 1.7 Systems of linear equations (HL)
The expression under a square root (or even root) is negative, or a denominator equals zero — both make $f(x)$ undefined.	The vertical line test: if any vertical line crosses the graph more than once, the relation is not a function, since one input would map to more than one output.	Adding: $3x=6$, $x=2$, then $y=5-2=3$.
IB-MATHAA-2.1-2ANSWER	IB-MATHAA-2.1-1ANSWER	IB-MATHAA-1.7-4ANSWER
Mathematics AA 2.2 Composite and inverse functions	Mathematics AA 2.1 Functions, domain and range	Mathematics AA 2.1 Functions, domain and range
They are reflections of each other in the line $y=x$.	Need $x+4 \geq 0$, so domain is $x \geq -4$.	A. Squares are never negative, and 0 is attained at $x=0$, so range is $f(x) \geq 0$.
IB-MATHAA-2.2-1ANSWER	IB-MATHAA-2.1-4ANSWER	IB-MATHAA-2.1-3ANSWER
Mathematics AA 2.2 Composite and inverse functions	Mathematics AA 2.2 Composite and inverse functions	Mathematics AA 2.2 Composite and inverse functions
$g(1)=4$, $f(4)=8$.	B. Swapping and solving $y=x+5$ gives $f^{-1}(x)=x-5$.	If two different x -values give the same output, the inverse would have to map that one output value to two different inputs, which violates the definition of a function.
IB-MATHAA-2.2-4ANSWER	IB-MATHAA-2.2-3ANSWER	IB-MATHAA-2.2-2ANSWER
Mathematics AA 2.3 Quadratic functions and the discriminant	Mathematics AA 2.3 Quadratic functions and the discriminant	Mathematics AA 2.3 Quadratic functions and the discriminant
C. $\Delta=16-20=-4<0$, so no real roots.	The vertex has x -coordinate $x=-b/(2a)$; substitute this back into $f(x)$ to find the y -coordinate.	$\Delta=b^2-4ac$. $\Delta>0$: two distinct real roots; $\Delta=0$: one repeated real root; $\Delta<0$: no real roots (complex conjugate pair).
IB-MATHAA-2.3-3ANSWER	IB-MATHAA-2.3-2ANSWER	IB-MATHAA-2.3-1ANSWER
Mathematics AA 2.4 Transformations of graphs	Mathematics AA 2.4 Transformations of graphs	Mathematics AA 2.3 Quadratic functions and the discriminant
It compresses the graph vertically towards the x -axis by factor $ a $ and reflects it in the x -axis.	$y=f(x)+d$ translates the graph vertically by d (up if $d>0$). $y=f(x+c)$ translates it horizontally by $-c$ (i.e. c units left if $c>0$).	$x-(-2)/2=1$, $f(1)=1-2+4=3$. Vertex $(1,3)$.
IB-MATHAA-2.4-2ANSWER	IB-MATHAA-2.4-1ANSWER	IB-MATHAA-2.3-4ANSWER



Mathematics AA 2.5 Exponential and logarithmic functions	Mathematics AA 2.4 Transformations of graphs	Mathematics AA 2.4 Transformations of graphs
Domain $x > 0$, range all real numbers, vertical asymptote $x = 0$ (no horizontal asymptote).	(1, 3), since the graph is translated down 2 units.	B. Multiplying the output by 3 stretches the graph vertically by factor 3.
IB-MATHAA-2.5-1ANSWER	IB-MATHAA-2.4-4ANSWER	IB-MATHAA-2.4-3ANSWER
Mathematics AA 2.5 Exponential and logarithmic functions	Mathematics AA 2.5 Exponential and logarithmic functions	Mathematics AA 2.5 Exponential and logarithmic functions
As $x \rightarrow \infty$, $0.5^x \rightarrow 0$, so $y \rightarrow -4$: asymptote $y = -4$.	C. $4^0 + 1 = 1 + 1 = 2$.	Because $\ln(x)$ is defined as the inverse function of e^x , and the graph of any function's inverse is always the reflection of the original graph in $y = x$.
IB-MATHAA-2.5-4ANSWER	IB-MATHAA-2.5-3ANSWER	IB-MATHAA-2.5-2ANSWER
Mathematics AA 2.6 Rational functions (HL)	Mathematics AA 2.6 Rational functions (HL)	Mathematics AA 2.6 Rational functions (HL)
B. Denominator zero: $2x - 6 = 0$, $x = 3$.	Because rewriting it as $A + B/(cx + d)$ ($B \neq 0$) shows the fraction term $B/(cx + d)$ is never exactly zero for finite x , so $y \neq A$ for any real x .	Vertical asymptote: solve $cx + d = 0$ for x . Horizontal asymptote: $y = a/c$, the ratio of the leading coefficients (from long division as $x \rightarrow \pm \infty$).
IB-MATHAA-2.6-3ANSWER	IB-MATHAA-2.6-2ANSWER	IB-MATHAA-2.6-1ANSWER
Mathematics AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics AA 2.6 Rational functions (HL)
The remainder theorem gives the remainder $P(k)$ when $P(x)$ is divided by $(x - k)$ for ANY k (which may be nonzero); the factor theorem is the special case where that remainder equals zero, showing $(x - k)$ divides $P(x)$ exactly.	$(x - k)$ is a factor of polynomial $P(x)$ if and only if $P(k) = 0$.	$y = 4/2 = 2$.
IB-MATHAA-2.7-2ANSWER	IB-MATHAA-2.7-1ANSWER	IB-MATHAA-2.6-4ANSWER
Mathematics AA 3.1 Volume and surface area of 3D solids	Mathematics AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics AA 2.7 Polynomial functions and factor theorem (HL)
Cone: $V = (1/3)\pi r^2 h$. Sphere: $V = (4/3)\pi r^3$.	By the remainder theorem, remainder = $P(2) = 8 - 4 + 1 = 5$.	C. Since $P(2) = 0$, by the factor theorem $(x - 2)$ is a factor.
IB-MATHAA-3.1-1ANSWER	IB-MATHAA-2.7-4ANSWER	IB-MATHAA-2.7-3ANSWER
Mathematics AA 3.1 Volume and surface area of 3D solids	Mathematics AA 3.1 Volume and surface area of 3D solids	Mathematics AA 3.1 Volume and surface area of 3D solids
$V = 3 \times 4 \times 5 = 60 \text{ cm}^3$.	B. Curved surface area = $2\pi rh = 2\pi(2)(7) = 28\pi$.	The slant height l is the distance from the apex to the edge of the base circle, found using Pythagoras: $l = \sqrt{r^2 + h^2}$, where r is the base radius and h the vertical height.
IB-MATHAA-3.1-4ANSWER	IB-MATHAA-3.1-3ANSWER	IB-MATHAA-3.1-2ANSWER
Mathematics AA 3.2 Right-angled trigonometry and bearings	Mathematics AA 3.2 Right-angled trigonometry and bearings	Mathematics AA 3.2 Right-angled trigonometry and bearings
B. They are equal, being alternate angles formed by the horizontal lines and the line of sight.	Bearings are measured clockwise from north, and always written as three digits (e.g. 045° , 302°), ranging from 000° to 360° .	$\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta = \text{adjacent/hypotenuse}$, $\tan \theta = \text{opposite/adjacent}$.
IB-MATHAA-3.2-3ANSWER	IB-MATHAA-3.2-2ANSWER	IB-MATHAA-3.2-1ANSWER



Mathematics AA 3.3 Non-right triangles: sine and cosine rules	Mathematics AA 3.3 Non-right triangles: sine and cosine rules	Mathematics AA 3.2 Right-angled trigonometry and bearings
It arises when given two sides and a non-included angle (SSA); after finding one possible angle via arcsin, check if 180° minus that angle also gives a valid triangle (angle sum <180° and consistent with given info).	Sine rule: $a/\sin A=b/\sin B=c/\sin C$. Cosine rule: $a^2=b^2+c^2-2bc\cdot\cos A$.	height = $6 \sin(60^\circ) = 6(0.8660) = 5.20$ m.
IB-MATHAA-3.3-2ANSWER	IB-MATHAA-3.3-1ANSWER	IB-MATHAA-3.2-4ANSWER
Mathematics AA 3.4 The unit circle and radian measure	Mathematics AA 3.3 Non-right triangles: sine and cosine rules	Mathematics AA 3.3 Non-right triangles: sine and cosine rules
Arc length $s=r\theta$; sector area $A=(1/2)r^2\theta$.	$c^2=36+36-2(6)(6)\cos 60^\circ=72-72(0.5)=36$ $c=6$.	A. $\text{Area}=(1/2)(6)(8)\sin 90^\circ=(1/2)(48)(1)=24$.
IB-MATHAA-3.4-1ANSWER	IB-MATHAA-3.3-4ANSWER	IB-MATHAA-3.3-3ANSWER
Mathematics AA 3.4 The unit circle and radian measure	Mathematics AA 3.4 The unit circle and radian measure	Mathematics AA 3.4 The unit circle and radian measure
$s=r\theta=5(\pi/2)=5\pi/2 \approx 7.85$ cm.	C. $\pi/3 \times 180/\pi = 60^\circ$.	For angle θ measured anticlockwise from the positive x-axis, $\cos\theta$ and $\sin\theta$ are respectively the x-coordinate and y-coordinate of the point where the terminal ray meets the unit circle.
IB-MATHAA-3.4-4ANSWER	IB-MATHAA-3.4-3ANSWER	IB-MATHAA-3.4-2ANSWER
Mathematics AA 3.5 Trigonometric identities and equations	Mathematics AA 3.5 Trigonometric identities and equations	Mathematics AA 3.5 Trigonometric identities and equations
B. This is the Pythagorean identity, true for all x.	Because dividing the domain by 2 too early loses solutions; you must first find all values of $2x$ in the extended domain $[0, 2\times\text{upper bound}]$, then divide each by 2 to get x, ensuring no solutions for x are missed.	$\sin^2\theta+\cos^2\theta=1$. $\cos(2\theta) = \cos^2\theta-\sin^2\theta = 2\cos^2\theta-1 = 1-2\sin^2\theta$ (all equivalent forms).
IB-MATHAA-3.5-3ANSWER	IB-MATHAA-3.5-2ANSWER	IB-MATHAA-3.5-1ANSWER
Mathematics AA 3.6 Vectors and scalar/vector products (HL)	Mathematics AA 3.6 Vectors and scalar/vector products (HL)	Mathematics AA 3.5 Trigonometric identities and equations
$ a \times b $ equals the area of the parallelogram formed by a and b; the area of the triangle formed by those two vectors as sides is half of this, $(1/2) a \times b $.	$a \cdot b = a b \cos\theta$, where θ is the angle between the vectors.	$x=\pi/6$ or $x=\pi-\pi/6=5\pi/6$.
IB-MATHAA-3.6-2ANSWER	IB-MATHAA-3.6-1ANSWER	IB-MATHAA-3.5-4ANSWER
Mathematics AA 3.7 Lines and planes in three dimensions (HL)	Mathematics AA 3.6 Vectors and scalar/vector products (HL)	Mathematics AA 3.6 Vectors and scalar/vector products (HL)
Line: $r = a + tb$, where a is a position vector to a known point on the line and b is a direction vector, t a parameter. Plane: $n \cdot r = n \cdot a$ (equivalently $ax+by+cz=d$), where $n=(a,b,c)$ is a normal vector to the plane and the point a lies on it.	$ v =\sqrt{(9+16+0)}=\sqrt{25}=5$.	B. $a \cdot b = 1(0)+0(1)+0(0) = 0$, since they are perpendicular unit vectors.
IB-MATHAA-3.7-1ANSWER	IB-MATHAA-3.6-4ANSWER	IB-MATHAA-3.6-3ANSWER



Mathematics AA 3.7 Lines and planes in three dimensions (HL) Substitute: $t+t+t=6 \rightarrow 3t=6 \rightarrow t=2$, point (2,2,2). IB-MATHAA-3.7-4ANSWER	Mathematics AA 3.7 Lines and planes in three dimensions (HL) B. $n_1 \cdot n_2 = 0$, so the normals (and hence the planes) are perpendicular. IB-MATHAA-3.7-3ANSWER	Mathematics AA 3.7 Lines and planes in three dimensions (HL) Find the angle ϕ between the direction vector and the normal using $\cos\phi = d \cdot n /(d n)$, then the angle between the line and the plane is $90^\circ - \phi$. IB-MATHAA-3.7-2ANSWER
Mathematics AA 4.1 Sampling and data collection C. Simple random sampling gives each member an equal probability of being chosen. IB-MATHAA-4.1-3ANSWER	Mathematics AA 4.1 Sampling and data collection Sampling bias occurs when a sample systematically over- or under-represents parts of the population, e.g. from using convenience sampling, which excludes anyone not easily accessible to the researcher. IB-MATHAA-4.1-2ANSWER	Mathematics AA 4.1 Sampling and data collection The population is divided into distinct subgroups (strata), and a random sample is taken from each stratum proportional to its size, ensuring all subgroups are fairly represented. IB-MATHAA-4.1-1ANSWER
Mathematics AA 4.2 Statistical measures and displays The mean is sensitive to extreme values since it uses every data value in its calculation, while the median depends only on the middle value(s) and is therefore resistant (robust) to outliers. IB-MATHAA-4.2-2ANSWER	Mathematics AA 4.2 Statistical measures and displays $IQR = Q3 - Q1$. A value is typically considered an outlier if it is below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. IB-MATHAA-4.2-1ANSWER	Mathematics AA 4.1 Sampling and data collection Fraction $= 50/1000 = 1/20$. Women: $600/20 = 30$. IB-MATHAA-4.1-4ANSWER
Mathematics AA 4.3 Correlation and linear regression It measures the strength and direction of the linear relationship between two variables; r ranges from -1 (perfect negative linear correlation) to +1 (perfect positive linear correlation), with 0 meaning no linear correlation. IB-MATHAA-4.3-1ANSWER	Mathematics AA 4.2 Statistical measures and displays Range $= 14 - 3 = 11$. Median $= 8$; $Q1 = 5$, $Q3 = 11$; $IQR = 11 - 5 = 6$. IB-MATHAA-4.2-4ANSWER	Mathematics AA 4.2 Statistical measures and displays C. The line inside the box of a box plot represents the median. IB-MATHAA-4.2-3ANSWER
Mathematics AA 4.3 Correlation and linear regression $y = 3(4) + 2 = 14$. IB-MATHAA-4.3-4ANSWER	Mathematics AA 4.3 Correlation and linear regression C. r closest to 0 indicates the weakest linear relationship. IB-MATHAA-4.3-3ANSWER	Mathematics AA 4.3 Correlation and linear regression Because the linear model was fitted only to data within a certain range, there is no evidence the same linear relationship continues to hold outside that range, so predictions far beyond it may be inaccurate. IB-MATHAA-4.3-2ANSWER
Mathematics AA 4.4 Probability, Venn and tree diagrams A. Since $P(A) \times P(B) = 0.18 = P(A \cap B)$, the independence condition holds. IB-MATHAA-4.4-3ANSWER	Mathematics AA 4.4 Probability, Venn and tree diagrams A and B are independent if $P(A \cap B) = P(A) \times P(B)$, equivalently $P(A B) = P(A)$ (knowing B occurred does not change the probability of A). IB-MATHAA-4.4-2ANSWER	Mathematics AA 4.4 Probability, Venn and tree diagrams $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If A and B are mutually exclusive, $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$. IB-MATHAA-4.4-1ANSWER



Mathematics AA

4.5 Discrete random variables and the binomial distribution

A fixed number n of independent trials, each with only two outcomes (success/failure), and a constant probability p of success on each trial.

IB-MATHAA-4.5-2

ANSWER

Mathematics AA

4.5 Discrete random variables and the binomial distribution

$$E(X)=np, \text{ Var}(X)=np(1-p).$$

IB-MATHAA-4.5-1

ANSWER

Mathematics AA

4.4 Probability, Venn and tree diagrams

$$P(A \cup B)=0.4+0.5-0=0.9.$$

IB-MATHAA-4.4-4

ANSWER

Mathematics AA

4.6 The normal distribution

$$z = (x-\mu)/\sigma, \text{ where } \mu \text{ is the mean and } \sigma \text{ the standard deviation of } X.$$

IB-MATHAA-4.6-1

ANSWER

Mathematics AA

4.5 Discrete random variables and the binomial distribution

$$E(X)=5(0.5)=2.5.$$
$$\text{Var}(X)=5(0.5)(0.5)=1.25.$$

IB-MATHAA-4.5-4

ANSWER

Mathematics AA

4.5 Discrete random variables and the binomial distribution

C. The total probability across all possible outcomes must sum to 1.

IB-MATHAA-4.5-3

ANSWER

Mathematics AA

4.6 The normal distribution

$$z=(25-20)/5=1.$$

IB-MATHAA-4.6-4

ANSWER

Mathematics AA

4.6 The normal distribution

C. The normal distribution is symmetric about its mean, so exactly half the probability lies below μ .

IB-MATHAA-4.6-3

ANSWER

Mathematics AA

4.6 The normal distribution

Approximately 68% within 1σ , 95% within 2σ , and 99.7% within 3σ of the mean.

IB-MATHAA-4.6-2

ANSWER

Mathematics AA

4.7 Bayes' theorem (HL)

B. $P(A)$ is the prior probability, updated to the posterior $P(A|B)$ after observing evidence B.

IB-MATHAA-4.7-3

ANSWER

Mathematics AA

4.7 Bayes' theorem (HL)

It computes the denominator $P(B)$ by summing over all ways B can occur: $P(B) = P(B|A)P(A) + P(B|A')P(A')$, needed since $P(B)$ is usually not given directly.

IB-MATHAA-4.7-2

ANSWER

Mathematics AA

4.7 Bayes' theorem (HL)

$$P(A|B) = P(B|A) \cdot P(A) / P(B).$$

IB-MATHAA-4.7-1

ANSWER

Mathematics AA

5.1 Limits and the derivative from first principles

The gradient of the tangent line to the curve $y=f(x)$ at the point x , i.e. the instantaneous rate of change of f at that point.

IB-MATHAA-5.1-2

ANSWER

Mathematics AA

5.1 Limits and the derivative from first principles

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h)-f(x)]/h.$$

IB-MATHAA-5.1-1

ANSWER

Mathematics AA

4.7 Bayes' theorem (HL)

$$P(A|B)=P(B|A)P(A)/P(B)=(0.8 \times 0.25)/0.4=0.2.$$

IB-MATHAA-4.7-4

ANSWER

Mathematics AA

5.2 Differentiation rules and tangents/normals

Product rule: $d/dx(uv)=u'v+uv'$.
Quotient rule: $d/dx(u/v)=(u'v-uv')/v^2$.

IB-MATHAA-5.2-1

ANSWER

Mathematics AA

5.1 Limits and the derivative from first principles

$$[4(x+h)-4x]/h = 4h/h = 4, \text{ so } f'(x)=4.$$

IB-MATHAA-5.1-4

ANSWER

Mathematics AA

5.1 Limits and the derivative from first principles

B. $[f(x+h)-f(x)]/h = [5-5]/h = 0$ for all $h \neq 0$, so the limit is 0.

IB-MATHAA-5.1-3

ANSWER

Mathematics AA

5.2 Differentiation rules and tangents/normals

$$dy/dx=2x+3, \text{ at } x=2: 4+3=7.$$

IB-MATHAA-5.2-4

ANSWER

Mathematics AA

5.2 Differentiation rules and tangents/normals

C. Power rule: $d/dx(x^5)=5x^4$.

IB-MATHAA-5.2-3

ANSWER

Mathematics AA

5.2 Differentiation rules and tangents/normals

The normal is perpendicular to the tangent, so if the tangent gradient is m ($m \neq 0$), the normal gradient is $-1/m$.

IB-MATHAA-5.2-2

ANSWER



Mathematics AA 5.3 Optimisation and related rates	Mathematics AA 5.3 Optimisation and related rates	Mathematics AA 5.3 Optimisation and related rates
A. $f'(x)=2x-6=0 \rightarrow x=3$ (and $f''(x)=2>0$ confirms minimum).	It links the rate of change of one quantity to another through a shared variable, e.g. $dV/dt = (dV/dr) \times (dr/dt)$, allowing you to find an unknown rate from a known one and a known relationship between the quantities.	If $f''(a)>0$, $x=a$ is a local minimum; if $f''(a)<0$, $x=a$ is a local maximum; if $f''(a)=0$, the test is inconclusive and another method (e.g. sign change of f') is needed.
IB-MATHAA-5.3-3ANSWER	IB-MATHAA-5.3-2ANSWER	IB-MATHAA-5.3-1ANSWER
Mathematics AA 5.4 Kinematics using calculus	Mathematics AA 5.4 Kinematics using calculus	Mathematics AA 5.3 Optimisation and related rates
Set the velocity function $v(t)=0$ and solve for t ; these are the instants where the particle's motion momentarily stops (though it may not be a permanent stop).	$v(t)=ds/dt$, $a(t)=dv/dt=d^2s/dt^2$. Conversely, $v=\int a \, dt$ and $s=\int v \, dt$.	$f'(x)=2x-4=0 \rightarrow x=2$. $f''(x)=2>0$, so $x=2$ is a minimum.
IB-MATHAA-5.4-2ANSWER	IB-MATHAA-5.4-1ANSWER	IB-MATHAA-5.3-4ANSWER
Mathematics AA 5.5 Integration and areas under curves	Mathematics AA 5.4 Kinematics using calculus	Mathematics AA 5.4 Kinematics using calculus
$\int x^n \, dx = x^{(n+1)}/(n+1)+C$ ($n \neq -1$). Also $\int e^x \, dx = e^x + C$ and $\int 1/x \, dx = \ln x + C$.	$v(t)=2t-4$, at $t=3$: $6-4=2$ m/s.	B. $v(t)=\int 4 \, dt=4t+C$; $v(0)=2$ gives $C=2$, so $v(t)=4t+2$.
IB-MATHAA-5.5-1ANSWER	IB-MATHAA-5.4-4ANSWER	IB-MATHAA-5.4-3ANSWER
Mathematics AA 5.5 Integration and areas under curves	Mathematics AA 5.5 Integration and areas under curves	Mathematics AA 5.5 Integration and areas under curves
x^3-2x+C .	C. $\int 2x \, dx = x^2$, evaluated: $9-1=8$.	The area formula $\int (f(x)-g(x))dx$ requires $f(x) \geq g(x)$ on the interval to give a positive area; subtracting in the wrong order gives a negative value that must otherwise be corrected by taking the absolute value or swapping the functions.
IB-MATHAA-5.5-4ANSWER	IB-MATHAA-5.5-3ANSWER	IB-MATHAA-5.5-2ANSWER
Mathematics AA 5.6 Volumes of revolution (HL)	Mathematics AA 5.6 Volumes of revolution (HL)	Mathematics AA 5.6 Volumes of revolution (HL)
B. $V=\pi \int [0 \text{ to } 5] 4 \, dx = \pi(4 \times 5) = 20\pi$.	Because the solid is modelled as a stack of thin circular discs of radius y , and the area of a disc is πy^2 , so summing (integrating) πy^2 over the thickness dx gives the total volume.	$V = \pi \int [a \text{ to } b] y^2 \, dx = \pi \int [a \text{ to } b] [f(x)]^2 \, dx$.
IB-MATHAA-5.6-3ANSWER	IB-MATHAA-5.6-2ANSWER	IB-MATHAA-5.6-1ANSWER
Mathematics AA 5.7 Further differentiation: implicit and higher derivatives (HL)	Mathematics AA 5.7 Further differentiation: implicit and higher derivatives (HL)	Mathematics AA 5.6 Volumes of revolution (HL)
Because such equations cannot easily be rearranged into the explicit form $y=f(x)$ (or would require splitting into multiple branches), so differentiating both sides directly while treating y as an implicit function of x is more efficient.	Apply the chain rule, treating y as a function of x : $d/dx(y^2)=2y \cdot dy/dx$. For a product term like xy , use the product rule: $d/dx(xy)=y+x \cdot dy/dx$.	$V = \pi \int [0 \text{ to } 1] x^4 \, dx$.
IB-MATHAA-5.7-2ANSWER	IB-MATHAA-5.7-1ANSWER	IB-MATHAA-5.6-4ANSWER

**Mathematics AA**

5.8 Further integration: substitution and by parts (HL)

$$\int u \, dv = uv - \int v \, du.$$

IB-MATHAA-5.8-1

ANSWER

Mathematics AA

5.7 Further differentiation: implicit and higher derivatives (HL)

$$2x - dy/dx = 0, \text{ so } dy/dx = 2x.$$

IB-MATHAA-5.7-4

ANSWER

Mathematics AA

5.7 Further differentiation: implicit and higher derivatives (HL)

$$\text{A. Differentiating: } 1 + 2y(dy/dx) = 0, \\ \text{so } dy/dx = -1/(2y).$$

IB-MATHAA-5.7-3

ANSWER

Mathematics AA

5.8 Further integration: substitution and by parts (HL)

Let $u = x^2$, $du = 2x \, dx$. Integral becomes $\int \cos(u) du = \sin(u) + C = \sin(x^2) + C$.

IB-MATHAA-5.8-4

ANSWER

Mathematics AA

5.8 Further integration: substitution and by parts (HL)

B. Since $du = 3x^2 dx$ exactly matches the rest of the integrand, $u = x^3 + 1$ is the natural substitution.

IB-MATHAA-5.8-3

ANSWER

Mathematics AA

5.8 Further integration: substitution and by parts (HL)

You must convert dx into an expression in terms of du (using $du = g'(x)dx$), and ideally rewrite the entire integral purely in terms of u before integrating.

IB-MATHAA-5.8-2

ANSWER

Mathematics AA

5.9 Differential equations and Maclaurin series (HL)

B. $\sin(x) = x - x^3/6 + x^5/120 - \dots$, so the coefficient of x^3 is $-1/6$.

IB-MATHAA-5.9-3

ANSWER

Mathematics AA

5.9 Differential equations and Maclaurin series (HL)

$f(x) = f(0) + f'(0)x + f''(0)x^2/2! + f'''(0)x^3/3! + \dots$, using successive derivatives of f evaluated at $x=0$.

IB-MATHAA-5.9-2

ANSWER

Mathematics AA

5.9 Differential equations and Maclaurin series (HL)

Rearrange to $(1/g(y))dy = f(x)dx$, integrate both sides separately, then apply any given initial condition to determine the constant of integration.

IB-MATHAA-5.9-1

ANSWER

Mathematics AA

5.9 Differential equations and Maclaurin series (HL)

$(1/y)dy = 2dx \rightarrow \ln|y| = 2x + C \rightarrow y = Ae^{(2x)}$; $y(0) = 5$ gives $A = 5$, so $y = 5e^{(2x)}$.

IB-MATHAA-5.9-4

ANSWER